HIGH RAINFALL AGRO-ECOLOGICAL RESEARCH LABORATORY, SHIVAMOGGA

SOIL HEALTH & FERTILITY TEST CERTIFICATE

SAMPLE DATA — FOR PROJECT DEMONSTRATION ONLY (FICTIONAL TEST DATA)

Soil Sample ID:	DEMO-SHC-KA04	**Report Date:**	18-Oct-2026
Location / District:	Thirthahalli Taluk, Shivamogga District, Karnataka	**Analytical Scheme:**	ICAR-NBSS&LUP; Standard Protocol
Soil Classification:	Laterite Soil (Oxisol / Ultisol)	**Target Agro-Zone:**	Karnataka - Shivamogga Region

SOIL CHEMICAL AND PHYSICAL PARAMETERS

Parameter Name	Symbol	Observed Value	Unit	Reference Range	Rating / Interpretation
Available Nitrogen	**N**	**162.0**	kg/ha	280.0 - 560.0	Low (Deficient)
Available Phosphorus	**P**	**12.8**	kg/ha	23.0 - 56.0	Very Low (Severe Fixation)
Available Potassium	**K**	**118.0**	kg/ha	140.0 - 330.0	Low (Leached)
Soil Reaction (pH)	**pH**	**5.30**	pH scale	6.5 - 7.5	Strongly Acidic
Electrical Conductivity	**EC**	**0.14**	dS/m	< 1.00	Normal (Highly Leached)
Organic Carbon	**OC**	**0.88**	%	0.50 - 0.75	High (Forest Humus)

STANDARD ANALYTICAL READOUT (EXTRACTABLE KEY-VALUE STREAM):

Available Nitrogen (N): 162.0 kg/ha [Low (Deficient)]

Available Phosphorus (P): 12.8 kg/ha [Very Low (Severe Fixation)]

Available Potassium (K): 118.0 kg/ha [Low (Leached)]

Soil Reaction (pH): 5.30 [Strongly Acidic]

Electrical Conductivity (EC): 0.14 dS/m [Normal (Highly Leached)]

Organic Carbon (OC): 0.88 % [High (Forest Humus)]

SOIL HEALTH INTERPRETATION & NUTRIENT ADVISORY

Diagnostic Assessment: High rainfall laterite soil characterized by strong acidity and severe phosphorus fixation by aluminium and iron oxides. High organic matter from forest canopy but basic cations are heavily leached.

General Nutrient Management Guidance: Broadcast agricultural lime or dolomite @ 2.5 t/ha once every 3 years. Use Rock Phosphate (Mussoorie Phos) or bone meal for slow-release phosphorus. Supplement with Boron (1 kg/ha) and Zinc Sulphate (15 kg/ha). Ideal for arecanut, pepper, and ginger.

Soil Analyst: Demonstration Technical Officer

Quality Authorized by: Senior Agricultural Chemist

Status: Fictional Demonstration Data

ACADEMIC DEMONSTRATION DISCLAIMER: *This soil test report contains fictional demonstration data generated strictly for research, user interface validation, and OCR testing in the MannuMitra AI-Based Predictive Farming System. It is not an official laboratory certificate, does not represent physical soil sampling of any real farming parcel, and must not be used as official government documentation or for commercial agricultural transactions.*